

Coding Challenge

Write a program that finds the value whose **second occurrence** appears earliest in the sequence, and calculates the **distance** (number of positions) between its first and second occurrence.

Requirements

1. Find the value that repeats **first** (earliest second occurrence)
 2. Calculate the distance between its first and second occurrence
 3. Distance = index of second occurrence - index of first occurrence
 4. Return **value distance** if a repeat exists
 5. Return **NONE** if no value appears more than once
 6. Use zero-based indexing for distance calculation
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Distance Calculation

Formula

distance = index_of_second_occurrence - index_of_first_occurrence

Example

Array: [5, 1, 3, 4, 3, 5, 6]

Index: 0 1 2 3 4 5 6

Value 3:

- First occurrence: index 2
- Second occurrence: index 4
- Distance: $4 - 2 = 2$

Value 5:

- First occurrence: index 0
- Second occurrence: index 5
- Distance: $5 - 0 = 5$

Winner: 3 (second occurrence at index 4 comes before 5's at index 5)

Input Format

- **Line 1:** Integer **n** (number of elements)
 - **Line 2:** **n** space-separated integers
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Output Format

If a repeat exists:

value distance

- **value** = the value with earliest second occurrence
- **distance** = number of positions between first and second occurrence

If no repeat exists:

NONE

Examples

Example 1: Basic Case

Input:

7

5 1 3 4 3 5 6

Output:

3 2

Explanation:

- **Value 5:** First at index 0, second at index 5 → distance = 5
- **Value 3:** First at index 2, second at index 4 → distance = 2
- **Value 1:** No repeat
- **Value 4:** No repeat
- **Value 6:** No repeat

- **Winner:** Value 3 (second occurrence at index 4 is earliest)
 - **Distance:** $4 - 2 = 2$
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Example 2: No Repeats

Input:

5

1 2 3 4 5

Output:

NONE

Explanation:

- Every value appears exactly once
- No repeats found

Constraints

- $1 \leq n \leq 100,000$
- Values can be positive, negative, or zero
- Value range: $-1,000,000 \leq \text{value} \leq 1,000,000$